

1. Consider the following dataset containing labeled text documents.

Document ID	Text	Label
1	The match was thrilling, and the team won.	Sports
2	The stock market is experiencing a downturn.	Finance
3	The player scored an amazing goal.	Sports
4	Investors are concerned about rising inflation.	Finance
5	The coach praised the team's performance.	Sports
6	The economic report predicts slower growth.	Finance

Test set:

Document ID	Text	Class
T1	The team played a fantastic match.	?

Describe step-by-step how a Naive Bayes classifier calculates the probability of a document belonging to either the Sports or Finance class. Use the document T1 ("The team played a fantastic match.") to illustrate your explanation.

Ans:

1) Preprocess

Lowercase & remove punctuation : ["the", "team", "played", "a", "fantastic", "match"]

2) Get conditional probabilities $P(w | C)$ and priors $P(C)$

Priors:

$$P(\text{Sports}) = 0.5, P(\text{Finance}) = 0.5$$

$P(w | \text{Sports})$:

- the = 0.1132
- team = 0.0566
- a = 0.0189
- match = 0.0377

$P(w | \text{Finance})$:

- the = 0.0588
- team = 0.0196
- a = 0.0392
- match = 0.0196

3) Compute the likelihood

- For **Sports**:

$$P(T1 | \text{Sports}) = 0.1132 \times 0.0566 \times 0.0189 \times 0.0189 \times 0.0189 \times 0.0377 \\ \approx 1.6307584 \times 10^{-9}$$

- For **Finance**:

$$P(T1 | \text{Finance}) = 0.0588 \times 0.0196 \times 0.0196 \times 0.0392 \times 0.0196 \times 0.0196 \\ \approx 3.4016347 \times 10^{-10}$$

4) Multiply by prior

- Sports numerator: $0.5 \times 1.6307584 \times 10^{-9} \approx 8.15379 \times 10^{-10}$
- Finance numerator: $0.5 \times 3.4016347 \times 10^{-10} \approx 1.70082 \times 10^{-10}$

5) Normalize to get $P(C | T1)$

$$P(C | T1) = \frac{\text{numerator}_C}{\sum_{C'} \text{numerator}_{C'}}$$

- Sum of numerators $\approx 9.85461 \times 10^{-10}$

So

- $P(\text{Sports} | T1) \approx \frac{8.15379 \times 10^{-10}}{9.85461 \times 10^{-10}} \approx 0.8274$
- $P(\text{Finance} | T1) \approx 1 - 0.8274 \approx 0.1726$

Thus, it can classify T1 as **Sports** (posterior $\approx 82.7\%$).

Attachments CPE393_Lab6.1 :

<https://docs.google.com/spreadsheets/d/1vu2dcRhgGGNU4E8ImwkJMRac6cfJFk7u6N04gRUGy9A/edit?gid=0#gid=0>